Potential Fields

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Where this fits:

- **Builds on:** Partial derivatives and gradients; Gradient descent; Local minima and saddle points; Distance queries and GJK.
- **Compare with:** Navigation function and feedback plan; Value iteration on a grid; Rapidly-exploring random tree (RRT).

1. Overview

Key point: A **potential field** (G-2740) gives every placement of the robot a height that is lowest at the goal and rises near obstacles; the robot reaches the goal by stepping downhill from wherever it is, with no grid and no search. The catch is a dent: a low spot that is not the goal, where every step goes uphill.

So far the robot finds its way to the door on a grid. The wavefront (G-2700) gives every cell its distance to the door, so stepping to the neighbour with the lowest number always arrives, and D* (G-2713) repairs those numbers when a box appears. What still breaks: every cell is numbered before the robot moves. A grid of points 0.05 m apart, fine enough for the robot's 0.25 m footprint, has 11,421 points in the room; a robot that also turns needs a number for every heading too.

This Note builds a **potential field** (G-2740): a height computed from where the robot is, from two things only: how far the door is, and how far the nearest obstacles are. The robot steps downhill, like a ball rolling on a landscape (MR §10.6.1). Figure 1 sends two robots of the disk footprint (radius 0.25 m) downhill on the same field at once. Watch the green robot from the start (2, 1) and the red one from (4.0, 0.5), between the table and the bottom wall, a second start chosen because from there the table stands between the robot and the door.

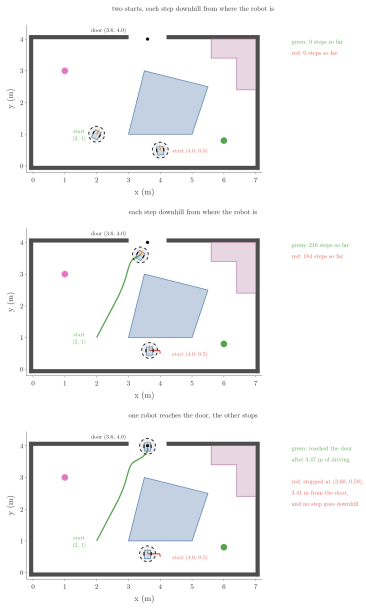


Figure 1: Animation: the room seen from above; two robots, drawn with their 0.25 m disk footprint, each step downhill on the same potential field. Green, from (2, 1): it rounds the table's corner and reaches the doorway centre (3.6, 4.0) after 3.47 m of driving. Red, from (4.0, 0.5): it slides left under the table and stops at (3.60, 0.59), 3.41 m from the door, where no step goes downhill. The right panel counts the steps

In the last frame, the green robot is at the door after 3.47 m; the red one has stopped at (3.60, 0.59), still 3.41 m from the door, although nothing blocks the way round the table. The opening question of this Note: **how can the robot pick each step from where it stands, why does the robot under the table stop, and how can it reach the door anyway?** We answer it in steps:

- the pull of the door, a bowl whose bottom is the door (Section 2);
- the push of the obstacles, a hill round each one (Section 3);
- why the robot under the table stops: a dent in the sum (Section 4);
- shaking the robot out of the dent at random (Section 5);
- a field built so that no dent can exist (Section 6);
- a formula with no dent that needs no grid, and where it works (Section 7).

Every number below is computed in the notebook `RO-050-potential-fields.ipynb`.

2. The pull of the door: the attractive potential

> **Key point:** The **attractive potential** (G-2741) is a bowl with its bottom at the door: half the squared distance to the door. Minus its gradient points straight at the door, so stepping downhill on the bowl alone drives the robot in a straight line.

The first part of the opening problem: which way should the robot at (2, 1) step, using only where it stands? Picture the floor bent into a bowl whose lowest point is the door. A ball

put anywhere on it rolls towards the door. Figure 2 draws that bowl over the room, rolls the robot down it, then tilts the camera up until the bowl becomes its contour map (G-468): each ring is one height, as in a map of a hill.

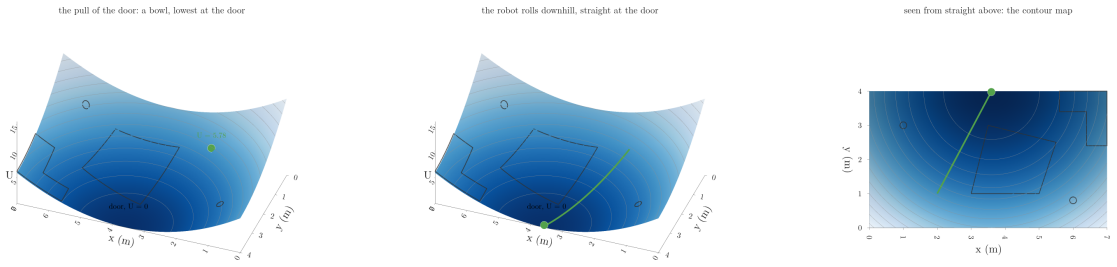


Figure 2: Animation: the attractive potential drawn as a surface over the room's floor, a bowl whose lowest point, 0, is the doorway centre (3.6, 4.0); the furniture is outlined on the surface only, because the bowl does not know it. The robot (green ball) at (2, 1), height 5.78, rolls straight down to the door. Then the camera tilts from the side to straight above: the surface becomes its contour map, rings of equal height round the door, with the straight path crossing them

The standard term. A height made from the distance to the goal is the **attractive potential** (G-2741) (MR §10.6.1; Choset slides). The symbols, on the robot at the start:

- q is the robot's configuration (G-2286), here the position (x, y) of its centre, in m: (2, 1) at the start; for the round robot the heading changes nothing;
- q_{goal} is the doorway centre, $(x_g, y_g) = (3.6, 4.0)$, in m;
- $\|q - q_{\text{goal}}\|$ is the straight-line distance from the robot to the door, in m: 3.4 m at the start;
- ζ (zeta) sets how steep the bowl is, 1 here, with no unit;
- $U_{\text{att}}(q)$ is the height at q, in m^2.

The height is half the squared distance to the door, times ζ:

$$U_{\text{att}}(q) = \frac{1}{2}\,\zeta\,\|q - q_{\text{goal}}\|^2$$

At the start, the robot is 1.6 m left of the door and 3.0 m below it:

$$\|q - q_{\text{goal}}\|^2 = 1.6^2 + 3.0^2$$

$$= 2.56 + 9 = 11.56 \text{ m}^2$$

$$U_{\text{att}} = \frac{1}{2} \times 11.56 = 5.78 \text{ m}^2$$

Which way is downhill. The gradient (G-863) of a height is the vector of its slopes along x and along y; it points the way the height rises fastest, and minus the gradient points the way it falls fastest (the gradient on the contour map). The slope along x, holding y at 1, by the power and chain rules, at the start:

$$U_{\text{att}} = \frac{1}{2}\Big((x - 3.6)^2 + 9\Big)$$

$$\frac{\partial U_{\text{att}}}{\partial x} = \frac{1}{2} \times 2\,(x - 3.6)$$

$$= x - 3.6$$

$$= 2 - 3.6 = -1.6 \text{ m}$$

The same steps along y, holding x at 2:

$$\frac{\partial U_{\text{att}}}{\partial y} = y - 4.0$$

$$= 1 - 4.0 = -3.0 \text{ m}$$

In symbols:

$$\frac{\partial U_{\text{att}}}{\partial x} = \zeta\,(x - x_g)$$

$$\frac{\partial U_{\text{att}}}{\partial y} = \zeta\,(y - y_g)$$

$$\nabla U_{\text{att}} = \zeta\,(q - q_{\text{goal}})$$

(Choset slides; MR §10.6.1.) At the start the gradient is $(-1.6, -3.0)$, so minus the gradient is (1.6, 3.0): from the robot straight to the door, as long as the distance, 3.4 m. The pull grows with the distance, like a spring stretched between the robot and the door (MR §10.6.1; Choset slides).

One step downhill. Stepping a small multiple of minus the gradient is **gradient descent** (G-862) (Choset slides; LaValle §8.4.1.2). With:

- α is the step factor, 0.01, with no unit; it plays the part of the learning rate (G-1068);
- ∇U is the gradient of the height at the robot, $(-1.6, -3.0)$ m at the start;
- q_{new} is the robot's next position, in m.

$$q_{\text{new}} = q - \alpha \nabla U$$

At the start:

$$q_{\text{new}} = (2, 1) - 0.01 \times (-1.6, -3.0)$$

$$= (2, 1) + (0.016, 0.030)$$

$$= (2.016, 1.030)$$

The step is 0.034 m long. We also cap every step at 0.05 m, because near an obstacle the slope gets very steep and an uncapped step could jump into it. Repeating the step until the robot is within 0.02 m of the door gives Figure 2: a straight line, the 3.4 m straight drive to the door, which happens to clear the table by 0.132 m.

> **Extra:** Here the gradient sets the robot's velocity. If it set the robot's acceleration instead, like a force on a ball with mass, the robot would roll past the door and back for ever, as a ball in a bowl without friction does; adding damping, a braking force that grows with speed, lets it settle (MR §10.6.1).

The bowl knows nothing about the furniture. From (4.0, 0.5), under the table, the straight line to the door runs into the table's grown band 0.25 m above the robot, and the bowl would drive the robot into it. The obstacles must change the height too.

3. The push of the obstacles: the repulsive potential

> **Key point:** The **repulsive potential** (G-2742) is a hill round each obstacle: zero beyond 0.5 m of clearance, then growing without bound as the clearance shrinks to zero. Added to the bowl, it bends the green robot's path round the table, 0.278 m clear of it; the red robot rolls under the table and stops.

The first part of the opening problem, continued: picking each step from where it stands, the robot under the table must not drive into the table. Picture a hill raised round every obstacle, steep close to it and flat a little way off. A ball rolling down the bowl then curves away from an obstacle it would hit.

The standard term. Each hill is a **repulsive potential** (G-2742) (MR §10.6.1; Choset slides). Each hill is computed from the robot's clearance to one obstacle, its distance query (G-2674) answer: the distance from the robot's body to that obstacle. The symbols, for one obstacle i:

- d_i is the robot's clearance to obstacle i, in m: the distance from the robot's centre to the obstacle minus the radius 0.25 m; at (4.0, 0.5), 0.25 m to the table and 0.25 m to the bottom wall;

- Q^* is the influence distance, 0.5 m: an obstacle farther than this does not push;
- η (eta) sets how strongly an obstacle pushes, 0.02, in m^4, so that the height comes out in m^2;
- $U_{\mathrm{rep},i}(q)$ is obstacle i's hill at q, in m^2.

$$U_{\mathrm{rep},i} = \frac{1}{2}\,\eta\left(\frac{1}{d_i} - \frac{1}{Q^*}\right)^2$$

for d_i up to Q^*, and 0 beyond (Choset slides).

Checks at easy values. At the edge of the influence distance, $d_i = 0.5$ m, the fraction $1/d_i$ equals $1/Q^*$, so the bracket is zero: the hill starts at height 0 with no step. At the robot under the table, $d_i = 0.25$ m:

$$U_{\mathrm{rep}} = \frac{1}{2} \times 0.02 \times (4-2)^2$$

$$= 0.04 \ \mathrm{m}^2$$

At 0.1 m the hill is 0.64 m^2, and as d_i shrinks to 0 the term $1/d_i$, and with it the hill, grows without bound: the robot can never roll into contact.

The push. The slope of the hill along the clearance, by the chain rule, at $d_i = 0.25$ m:

$$\frac{dU_{\mathrm{rep}}}{dd} = 0.02 \times (4-2) \times \left(-\frac{1}{0.25^2}\right)$$

$$= 0.04 \times (-16) = -0.64 \ \mathrm{m}$$

The same in symbols:

$$\frac{dU_{\mathrm{rep}}}{dd} = \eta\left(\frac{1}{d} - \frac{1}{Q^*}\right)\left(-\frac{1}{d^2}\right)$$

The clearance grows fastest straight away from the obstacle's nearest point, one metre of clearance per metre moved, so its gradient is the unit vector n_i pointing from that nearest point to the robot's centre. So:

$$\nabla U_{\mathrm{rep},i} = -\eta\left(\frac{1}{d_i} - \frac{1}{Q^*}\right)\frac{1}{d_i^2}\,n_i$$

(Choset slides.) Minus this gradient, the push, points along n_i, away from the obstacle, with size 0.64 at a clearance of 0.25 m.

The whole field. The robot's height U, in m^2, is the bowl plus one hill for each of the nine obstacles: the table, the cabinet, the two poles and five pieces of wall. The sign $\sum$ adds up the nine hills, for i from 1 to 9:

$$U = U_{\text{att}} + \sum_{i=1}^{9} U_{\text{rep},i}$$

and its gradient is the sum of their gradients (MR §10.6.1). At the start (2, 1) no obstacle is within 0.5 m, so the height is the bowl's 5.78 m^2 and the first step is that of Section 2. Under the table, at (4.0, 0.5), the bowl gives

$$U_{\text{att}} = \frac{1}{2}\,(0.4^2 + 3.5^2)$$

$$= \frac{1}{2} \times (0.16 + 12.25)$$

$$= 6.205 \text{ m}^2$$

and the table and the bottom wall each add 0.04 m^2:

$$U = 6.205 + 0.04 + 0.04 = 6.285 \text{ m}^2$$

The two pushes, 0.64 down from the table and 0.64 up from the wall, cancel, so the first step is the bowl's alone, up and a little left.

Figure 3 draws the whole field as a surface over the middle of the room and rolls both robots down it, then tilts the camera to the contour map. Watch the green robot's path bend away from the table's corner, and the red robot slide left along the strip under the table and stop.

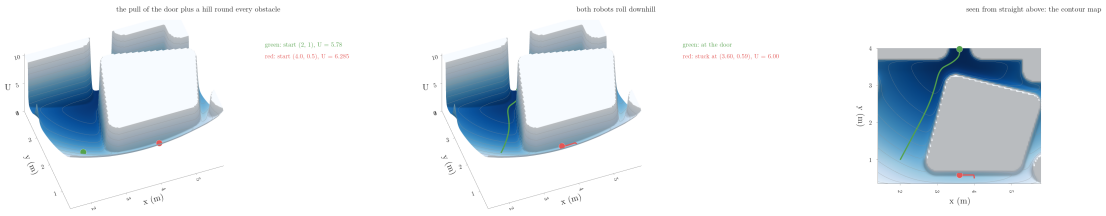


Figure 3: Animation: the total potential, the attractive bowl plus a repulsive hill round every obstacle, drawn as a surface over the middle of the room (x from 1.4 to 5.8 m, y from 0.35 to 4 m). Heights above 10 m^2 are cut off at 10, so every obstacle with its 0.25 m band shows as a flat-topped block. The green robot from (2, 1), height 5.78, rolls round the table's corner to the doorway; the red robot from (4.0, 0.5), height 6.285, rolls left under the table and stops at (3.60, 0.59), height 6.00. Then the camera tilts to the contour map with both paths

The green robot reaches the door after 705 steps and 3.47 m, never closer than 0.278 m to an obstacle, against 0.132 m on the straight drive: the table's hill pushes it away from the corner. Each step needed nine clearances and one subtraction for the bowl, not a number for every cell of the room. The red robot stops after 598 steps at (3.60, 0.59), 3.41 m from the door: the second part of the opening problem asks why.

4. Why the robot stops: local minima

> **Key point:** Under the table the door's pull, up, exactly balances the table's push, down, so the gradient is zero and every step goes uphill: a **local minimum** (G-1110) of the field. Every start in the strip under the table between x = 2.95 and 5.25 m, 7.7 percent of the starts tried, rolls into it.

The second part of the opening problem: why did the red robot of Figure 3 stop at (3.60, 0.59)? Because the gradient there is zero: the door's pull and the obstacles' pushes cancel. At that point:

- the robot is 3.407 m straight below the door, so the pull, minus the bowl's gradient, is 3.407 m, straight up (Section 2);
- its clearance to the table above is 0.15679 m, so the table pushes down;
- its clearance to the bottom wall is 0.3432 m, so the wall pushes up, a little.

The size of the table's push, minus its hill's gradient, by the formula of Section 3, in m:

$$0.02 \times \left(\frac{1}{0.15679} - 2 \right) \times \frac{1}{0.15679^2}$$

$$= 0.02 \times 4.378 \times 40.68$$

$$= 3.562$$

The wall's push, the same way with 0.3432 m, is 0.155 m. Up and down balance: the pull plus the wall's push equals the table's push, in m:

$$3.407 + 0.155 = 3.562$$

A picture of the balance. Figure 4 cuts the field along the line x = 3.6 m, from the bottom wall's band (y = 0.25 m) to the table's band (y = 0.75 m). The bowl falls as the robot moves up, towards the door; the table's hill rises as it nears the table. Their sum has a lowest point at y = 0.593 m, height 6.003 m^2: a dent.

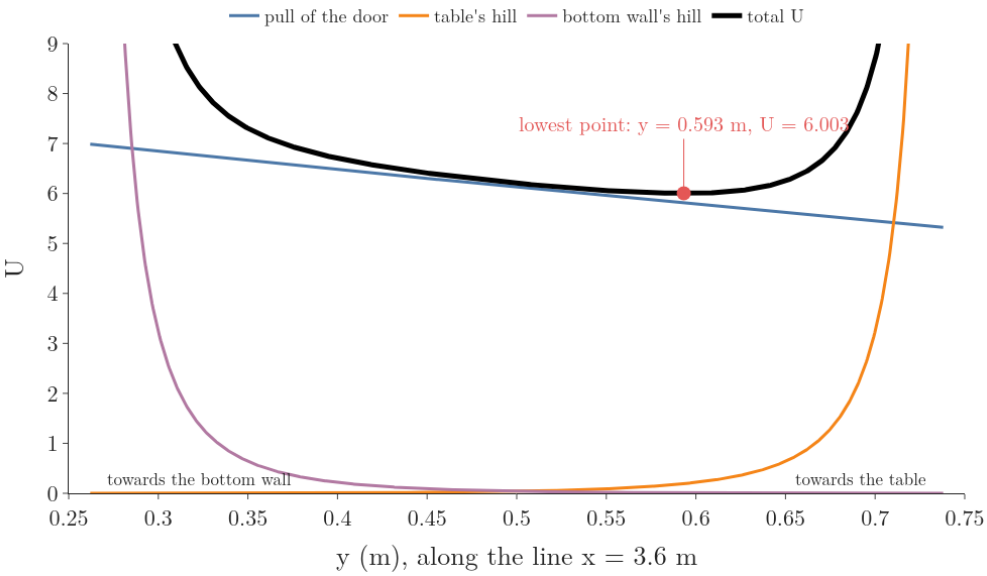


Figure 4: The potential along the line x = 3.6 m under the table, from y = 0.262 to 0.738 m: the pull of the door, the bowl (blue), falls from 7.0 to 5.3 m^2 as y grows; the table's hill (orange) rises steeply near the table, the bottom wall's hill (purple) near the wall; their sum, the total height U (black), has its lowest point at y = 0.593 m, U = 6.003 m^2 (red dot)

Along x the dent is a bowl too. The table's bottom edge is straight from x = 3.0 to 5.0 m, so its push is the same at every x below it, and only the door's pull changes with x: it is smallest straight below the door, at x = 3.6 m. The robot slid left until it got there.

The standard terms. A point lower than every point around it, but not the lowest point overall, is a **local minimum** (G-1110); the lowest point of all, here the door, is the **global minimum** (G-848). A point where the gradient is zero is a **stationary point** (G-1879), also called a critical point: a minimum, a maximum or a saddle (LaValle §8.4.4.3). The field has three **saddle points** (G-1718), stationary points that curve up one way and down another:

- (2.92, 0.60), at the left end of the strip under the table, height 6.206 m^2;
- (5.17, 0.63), at its right end, height 7.105 m^2;
- (0.52, 2.84), between the pink pole and the left wall, height 5.64 m^2.

The difference matters. A robot stops on a saddle only if it starts exactly on the line that runs into it, and the smallest nudge sends it on. A local minimum catches a whole region of starts, all those whose downhill paths end in it, because it attracts the points near it (MR §10.6.1).

Figure 5 shows that region. The animation starts 82 robots on a grid 0.4 m apart and lets them all roll downhill at once, then colours a start every 0.1 m by where it ends.

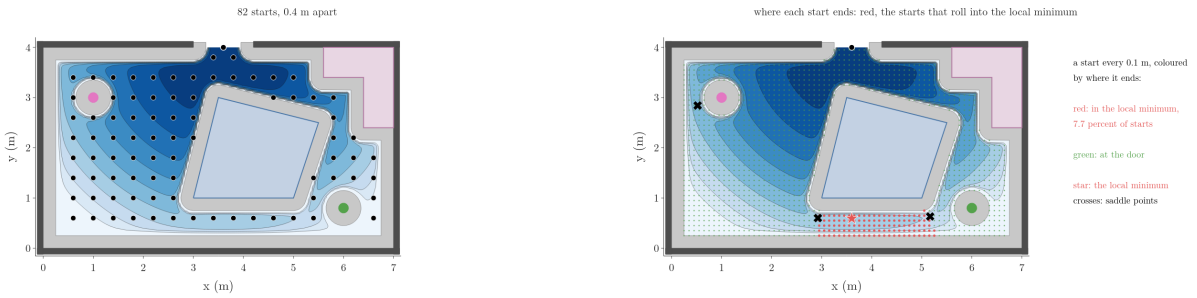


Figure 5: Animation: the contour map of the total potential over the whole room (colours and contour lines as in the last frame of Figure 3, heights above 10 m^2 drawn as 10; light grey: obstacles with their 0.25 m band). 82 robots start 0.4 m apart and all follow the gradient downhill at once; the six that start between the table and the bottom wall end in the local minimum at (3.60, 0.59), all the others at the door. The last frame colours a start every 0.1 m by where it ends: red, in the local minimum (7.7 percent of the 1,529 starts), green, at the door; the star is the local minimum and the black crosses are the three saddle points

In the last frame of Figure 5, the red region is the strip under the table between the two saddles, x from 2.95 to 5.25 m: 118 of the 1,529 starts, 7.7 percent. Every start there ends in the local minimum, and every other start reaches the door. Local minima like this one come from adding a bowl and hills: where an obstacle lies between the robot and the goal, the pull and the push can balance, and it is hard in general to build such a field without them (MR §10.6.2; LaValle §8.4.1.2). Gradient descent alone can never leave one, because leaving means going uphill first.

5. Escaping by chance: randomized potential fields

> **Key point:** A **randomized potential field** (G-2744) descends until stuck, then takes a **random walk** (G-2745), a string of small random steps, until it finds a point lower than the local minimum, and descends again. From under the table it always reached the door in 200 runs, but drove 46 m (median) where the shortest route is 4.55 m.

The third part of the opening problem: how can the robot under the table reach the door anyway? The robot sits in the local minimum at (3.60, 0.59), height 6.003 m^2, and every small step goes uphill. Picture a marble stuck in a hollow of a tilted tray: shaking the tray bounces it about until it happens to clear the rim, and then it rolls on downhill.

The standard terms. A path made of steps each chosen at random, with no memory of the last, is a **random walk** (G-2745). A planner that descends the potential, walks at random when stuck, then descends again is a **randomized potential field** (G-2744) planner (LaValle §5.4.3). The planner switches between three modes (LaValle §5.4.3, Figure 5.15):

1. **Descend:** gradient descent, as in Section 3, until the robot stops.
2. **Random walk:** each step moves x by 0.05 m up or down and y by 0.05 m up or down, each decided by a coin toss; a step into the obstacle region is thrown away and tried

again. The walk ends as soon as the height is below the local minimum's 6.003 m^2, or after a step limit drawn at random from 100 to 400 steps. Then back to Descend.

3. **Backtrack:** after 20 walks that all fall back into the same local minimum, restart from a point picked at random among the points of the earlier walks.

Figure 6 runs it from the local minimum. Watch walk 1 wander right and fall back, then walk 2.

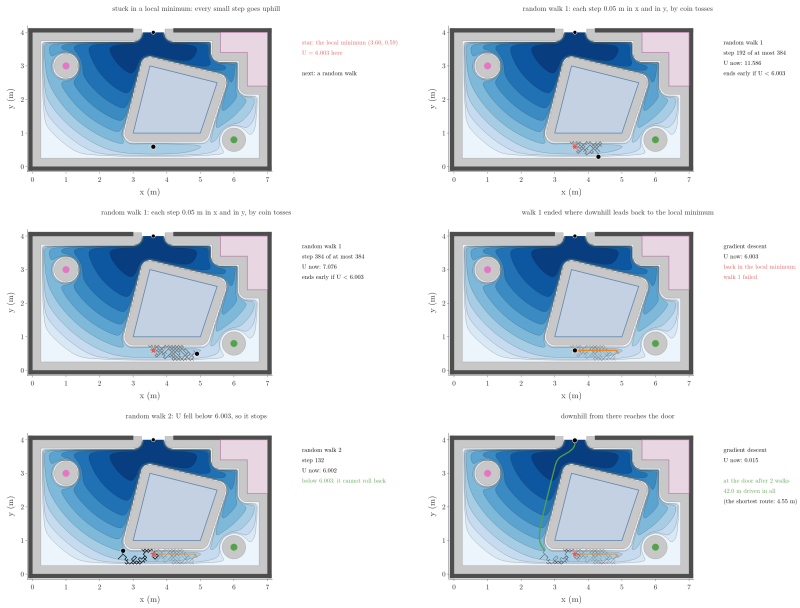


Figure 6: Animation: the randomized potential field on the contour map of Figure 5. Stuck in the local minimum at (3.60, 0.59), height 6.003 m^2 (red star). Random walk 1 (grey) takes 384 steps of 0.05 m in x and in y and ends at (4.90, 0.49), height 7.076, still among the starts that roll back into it; gradient descent (orange) rolls it back into the local minimum. Random walk 2 (black) stops after 132 steps at (2.70, 0.69), where the height, 6.002, is below 6.003; gradient descent from there (green) reaches the door. 42.0 m driven in all; the shortest route is 4.55 m

In Figure 6, walk 1 ended at (4.90, 0.49), inside the red region of Figure 5, so the descent fell back into the local minimum. Walk 2 ended at (2.70, 0.69), past the left saddle, where the height had dropped to 6.002 m^2, below the local minimum's 6.003. From there the descent of Section 3 reaches the door. The run drove 42.0 m.

Why it needs many steps. To leave, the walk must climb over a saddle, 0.68 m to the left of the local minimum or 1.57 m to the right. Each step moves x by 0.05 m one way or the other, a variance of 0.0025 m^2 per step, and independent errors add their variances, so after 200 steps the variance is 200 times 0.0025 m^2, and the typical distance the walk has wandered along x, its square root, in m:

$$\sqrt{200 \times 0.05^2} = 0.05 \times \sqrt{200}$$

$$= 0.05 \times 14.14 = 0.71 \text{ m}$$

The same in symbols, with:

- n is the number of steps in the walk, 200 above;
- s is one step's move along x, either way, 0.05 m.

$$\sqrt{n\, s^2} = s\, \sqrt{n}$$

Two hundred random steps typically wander about as far as the left saddle, and much of that wandering is back and forth.

Over 200 runs (notebook):

- every run reached the door;
- it took 3.8 random walks on average, at most 22, and 32 percent of runs escaped on the first walk;
- only one of the 200 runs needed a backtrack;
- the distance driven had a median of 46.1 m; one run in ten drove less than 16.3 m, one in ten more than 168.0 m;
- 90 percent escaped to the left, over the lower saddle (6.206 m^2 against 7.105 m^2).

The shortest route from under the table round the table to the door is 4.55 m (notebook, a visibility graph over the grown obstacles). So the random walks free the robot, at a price: long, jittery paths that must be smoothed before a robot drives them, and three settings, the step, the walk limit and the number of walks before backtracking, that must be tuned for each problem (LaValle §5.4.3). The randomized potential field solved problems with up to 31 degrees of freedom when it appeared (LaValle §5.4.3). Escaping by luck is still escaping after the robot got stuck; a better field would have no local minimum to get stuck in.

6. A field with no local minima: harmonic potentials

> **Key point:** Hold every obstacle at height 1 and the door at 0, and let every free point take the average of its neighbours: the result, a **harmonic potential** (G-2748), has no local minimum, because an average is never below all its neighbours. Downhill from under the table reaches the door, but the heights must be solved on a grid of the whole room first.

The local minimum under the table came from adding a bowl and a hill; the random walks of Section 5 left it only by luck, after 46 m of driving. The third part of the opening problem, without luck: build a field with no local minimum at all. Picture the room as a metal plate whose obstacles are held hot and whose door is held cold. Heat flows from hot to cold until the temperatures stop changing. A cold spot inside the plate would draw heat from all sides and warm up, so in the settled plate the coldest place is the door, and from every point there is a way to somewhere colder.

One dimension first. Figure 7 does this on a corridor of 11 points 0.1 m apart, the left end against an obstacle held at 1, the right end the door held at 0. The 9 points between start at 1. A sweep visits them in turn and sets each to the average of its two neighbours' latest values. Watch the first sweep, point by point: point 9 becomes the average of 1 and 0, 0.5.

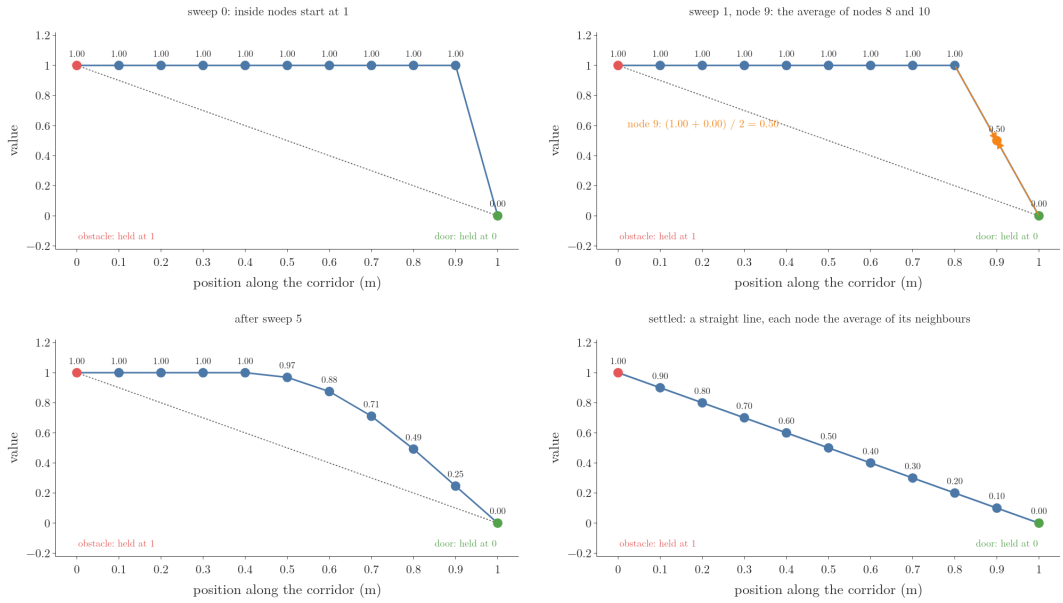


Figure 7: Animation: relaxation on a corridor of 11 points 0.1 m apart; the left point (red) is held at 1, the door (green) at 0, the 9 points between start at 1. Sweep 1 is shown point by point, each set to the average of its two neighbours: points 1 to 8 stay 1, point 9 becomes (1.00 + 0.00) / 2 = 0.50. Then whole sweeps: after sweep 5 the values are 1.00, . . . , 0.97, 0.88, 0.71, 0.49, 0.25, 0.00; after about 80 sweeps they settle on the straight line 0.9, 0.8, . . . , 0.1 (dotted)

The settled values in Figure 7 lie on a straight line, and on a straight line every point is the average of its two neighbours. No point is below both of its neighbours, so from every point a step towards the door goes downhill.

In two dimensions each point has four neighbours: left, right, below and above. Figure 8 shows one point of the room's settled field, at (3.5, 3.6), just below the door: its height is the average of its four neighbours.

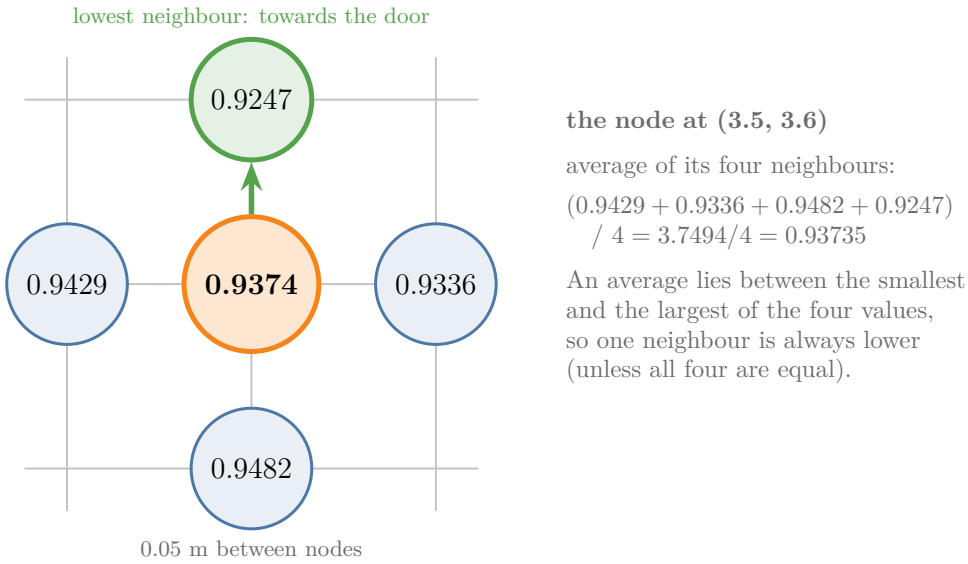


Figure 8: One free point of the room's settled harmonic potential, at (3.5, 3.6), and its four neighbours 0.05 m away: left 0.9429, right 0.9336, below 0.9482, above 0.9247; the point itself holds 0.9374, the average, 3.7494 / 4 = 0.93735. The lowest neighbour, above, towards the door, is the robot's next step (green arrow)

The standard terms. The settled temperature obeys **Laplace's equation** (G-2746): its bend along x plus its bend along y is zero everywhere inside (the sum of the two second partial derivatives, the Laplacian, is zero). A function that obeys it is a **harmonic function** (G-2747), and a harmonic function used as the robot's field, held at 1 on the obstacles and 0 at the goal, is a **harmonic potential** (G-2748) (LaValle §8.4.4.4; Wikipedia, Harmonic function).

From Laplace's equation to the average. On a grid with spacing h, the bend along x at a point is how much its two neighbours along x rise above it, added and divided by h^2, and the same along y (Wikipedia, Discrete Laplace operator). Laplace's equation sets the bend along x plus the bend along y to zero; multiplied by h^2, it says the four rises add up to zero. At (3.5, 3.6), with its height ϕ (phi) not yet known and its neighbours' heights 0.9429, 0.9336, 0.9482 and 0.9247:

$$(0.9429 - \phi) + (0.9336 - \phi)$$

$$+(0.9482 - \phi) + (0.9247 - \phi) = 0$$

$$3.7494 - 4\phi = 0$$

$$\phi = 3.7494/4 = 0.93735$$

which rounds to the point's settled 0.9374. The same steps in symbols, with:

- ϕ is the harmonic potential's height at the point, with no unit, 0.9374 at (3.5, 3.6);
- ϕ_L, ϕ_R, ϕ_D, ϕ_U are the heights of its left, right, lower and upper neighbours, 0.9429, 0.9336, 0.9482 and 0.9247;
- h is the grid spacing, 0.05 m.

$$\frac{(\phi_L - \phi) + (\phi_R - \phi)}{h^2}$$

$$+\frac{(\phi_D - \phi) + (\phi_U - \phi)}{h^2} = 0$$

$$\phi_L + \phi_R + \phi_D + \phi_U - 4\phi = 0$$

$$\phi = \frac{\phi_L + \phi_R + \phi_D + \phi_U}{4}$$

Why there is no local minimum. An average of four numbers is never below the smallest of them, and equals it only if all four are equal. So a point equal to the average of its neighbours always has a neighbour at least as low, and a strictly lower one unless all four are the same. A local minimum, a point lower than all its neighbours, cannot exist; the lowest value is on the boundary, at the door. No interior minimum is the minimum principle of harmonic functions (Wikipedia, Harmonic function; LaValle §8.4.4.4). In the room, the notebook checks it: of the 5,839 free points other than the door, every one has a strictly lower neighbour.

Computing the field. We sweep the grid, as in Figure 7, setting each point to the average of its neighbours' latest values, again and again; LaValle (§8.4.4.4) notes that such sweeps are related to value iteration (G-2704), which also sweeps until nothing changes. The room's grid, 0.05 m apart, has 11,421 points, 5,840 of them free. Starting from 1 everywhere but the door, the largest error after each number of sweeps is (notebook):

Sweeps	1	100	1,000	5,000
Largest error	0.33	6.2×10^{-3}	1.9×10^{-5}	4.2×10^{-11}

Figure 9 sends both robots downhill on the settled field: each step goes to the lowest of the 8 neighbouring points. The right panel shows the height at each robot.

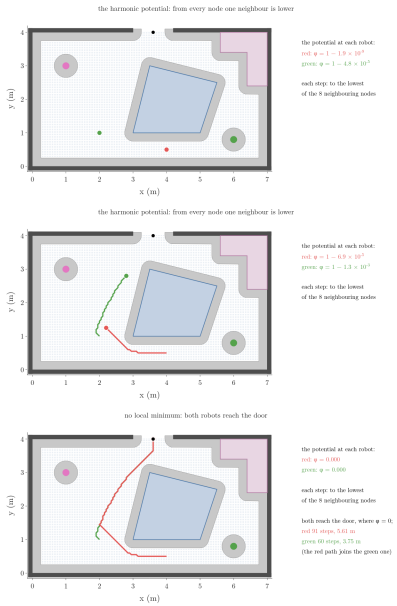


Figure 9: Animation: the room's grid of free points 0.05 m apart (light dots; light grey: obstacles with their 0.25 m band). Both robots step to the lowest of their 8 neighbouring points on the settled harmonic potential. The red robot from (4.0, 0.5), height $1 - 1.9 \times 10^{-9}$, rounds the table's left side and reaches the door in 91 steps, 5.61 m; the green robot from (2, 1), height $1 - 4.8 \times 10^{-5}$, in 60 steps, 3.75 m; the red path joins the green one. The panel shows the harmonic potential's height ϕ at each robot

The red robot that got stuck in Sections 3 and 4 now reaches the door, in 91 steps and 5.61 m, with no random walk: the third part of the opening problem is solved, on a grid. Two costs come with it:

- **The grid is back.** Before the robot moves, 5,000 sweeps over 5,839 points, 29.2 million updates, must settle the field; in more dimensions the grid explodes, and the boundary of free space must be known (LaValle §8.4.4.4).
- **The field is very flat.** Under the table the height is $1 - 1.9 \times 10^{-9}$, and at the highest free point of the room it is $1 - 2.3 \times 10^{-14}$: neighbouring heights differ only in the ninth to fourteenth decimal place, so the sweeps must run until the error is far smaller than that, and the computer must keep about 14 digits.

The robot now has a field with no local minimum, but only on a grid. What is left is a field with no local minimum given by a formula, computed from where the robot is, as in Section 3.

7. One minimum by design: navigation functions

> **Key point:** A **navigation function** (G-2750) in Rimon and Koditschek's sense is a smooth field with a single minimum, at the goal, and the same top value, 1, on every obstacle's edge. For a world of disks, a disk-shaped floor with disk-shaped obstacles, a formula with a knob κ gives one: with $\kappa = 1$ the robot under the table still stops in a local minimum; from $\kappa = 2$ it reaches the door.

The third part of the opening problem, without a grid: the harmonic field of Section 6 has no local minimum but needs the whole grid, while the bowl and hills of Section 3 need no grid but have a local minimum. A navigation function (G-2702) on the grid, the wavefront's numbers, already has the property we want: 0 at the goal and a lower neighbour everywhere else. Rimon and Koditschek asked for the same in a smooth field over free space, written as one formula.

The standard term. A **navigation function** (G-2750) in the sense of Rimon and Koditschek is a field on free space that (LaValle §8.4.4.3; Choset slides):

1. is smooth, so it has a gradient everywhere;
2. has only one minimum, at the goal;
3. takes the same, highest value, 1, on the whole boundary of free space, so that every obstacle's edge is equally high and the robot never reaches one;
4. has no flat stationary points: each one curves up or down in every direction, so none is a plateau where gradient descent cannot pick a direction.

Rule 2 does not forbid saddle points. In free space with holes, such as a room with furniture, a smooth field must have stationary points besides the goal (LaValle §8.4.4.3); saddles are harmless, because they catch the robot only from starts exactly on the lines that run into them.

A world made of disks. Rimon and Koditschek gave a formula for such a world: free space that is a disk (a ball in more dimensions) with disk-shaped obstacles inside it (Choset slides; LaValle §8.4.4.3). Our room is not one, so we redraw it as one, keeping the start under the table and the door:

- the floor the robot's centre may use is a disk of radius 3.3 m about (3.5, 2.0); it reaches 1.3 m below the room's bottom wall, so this world is a stand-in, not the room;
- the table, grown by the robot's 0.25 m, is a disk of radius 1.3 m about (4.2, 1.9), close to the grown table's size;
- the two poles, grown by 0.25 m, are disks of radius 0.35 m about (1, 3) and (6, 0.8).

The formula. The symbols (Choset slides):

- γ (gamma) is the squared distance from the robot to the door, in m^2: at (4.0, 0.5), 12.41 m^2;
- β_0 is the room disk's term, its radius squared minus the robot's squared distance from its centre, in m^2: 8.39 at (4.0, 0.5); it is 0 on the disk's edge and positive inside;
- β_i is obstacle i's term, the robot's squared distance from its centre minus its radius

squared, in m^2: 0.31 for the table, 15.128 for the pink pole and 3.968 for the green pole at (4.0, 0.5); it is 0 on the obstacle's edge and positive outside;
- β is the product of all four, 156.1 m^8 at (4.0, 0.5): 0 on every boundary of free space, positive inside it;
- κ (kappa) is a knob, a number of 1 or more, 2 here;
- ϕ is now the navigation function's height, with no unit; like the harmonic potential of Section 6 it is 0 at the goal and 1 on every boundary.

$$\phi = \frac{\gamma}{\left(\gamma^{\kappa} + \beta\right)^{1/\kappa}}$$

Checks at easy values. At the door, $\gamma = 0$, so $\phi = 0$: the lowest value. On any boundary, $\beta = 0$, so:

$$\phi = \frac{\gamma}{(\gamma^{\kappa})^{1/\kappa}} = \frac{\gamma}{\gamma} = 1$$

Inside free space β is positive, so the bottom of the fraction is larger than γ and ϕ lies between 0 and 1. At the robot under the table, with $\kappa = 2$:

$$\phi = \frac{12.41}{\sqrt{12.41^2 + 156.1}}$$

$$= \frac{12.41}{\sqrt{154.0 + 156.1}}$$

$$= 12.41/17.61 = 0.705$$

Figure 10 draws this field as a surface, rolls the robot from under the table down it, then tilts to the contour map.

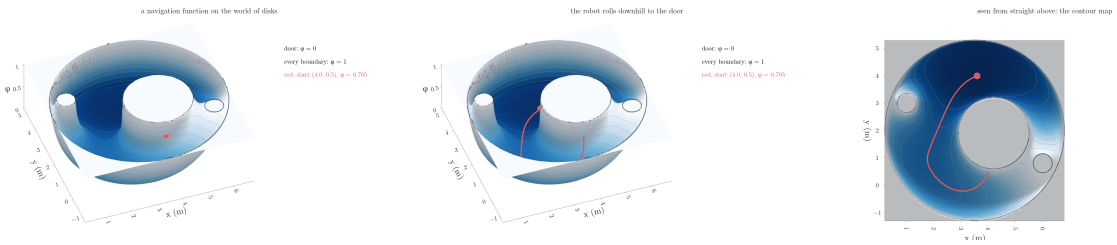


Figure 10: Animation: the Rimon–Koditschek navigation function, with its knob $\kappa = 2$, on the world of disks, drawn as a surface: the free disk of radius 3.3 m about (3.5, 2.0), the table as a disk of radius 1.3 m about (4.2, 1.9) and the two poles as disks of radius 0.35 m; heights run from 0 at the doorway centre (3.6, 4.0) to 1 on every boundary, and the area outside the free disk is drawn at 1. The red robot from (4.0, 0.5), height 0.705, rolls round the bottom of the table to the door. Then the camera tilts to the contour map

Why κ must be large enough. For small κ the obstacle term β dominates the bottom of the fraction and the formula has local minima of its own; for κ large enough the distance term wins and the only minimum left is the door (Choset slides, "for large enough κ, this is a navigation function"). Figure 11 raises κ step by step, with the robot from under the table each time.

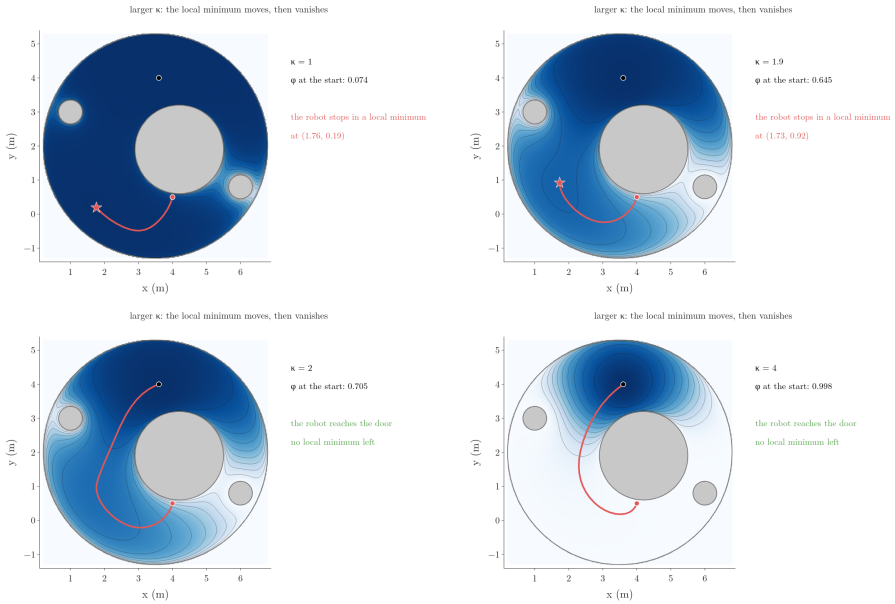


Figure 11: Animation: the contour map of the navigation function on the world of disks (as in the last frame of Figure 10) for the formula's knob $\kappa = 1$, 1.25, 1.5, 1.75, 1.9, 2, 3 and 4, with the path of the robot from (4.0, 0.5). For $\kappa = 1$ to 1.9 the robot stops in a local minimum left of the table (red star), at (1.76, 0.19) for $\kappa = 1$ and (1.73, 0.92) for $\kappa = 1.9$; from $\kappa = 2$ the local minimum is gone and it reaches the door. The height at the start grows with κ: 0.074 at $\kappa = 1$, 0.705 at $\kappa = 2$, 0.998 at $\kappa = 4$

In Figure 11 the local minimum moves up as κ grows and vanishes between 1.98 and 2 (notebook: the smallest κ that reaches the door from (4.0, 0.5) is 1.983). At $\kappa = 2$ the field has one minimum, the door, and three saddle points, one behind each obstacle, as rule 2 allows (notebook). The robot under the table reaches the door, which answers the third part of the opening problem without a grid, for the world of disks. Two limits remain:

- **Larger κ flattens the field.** The height at the start is 0.074 for $\kappa = 1$, 0.705 for $\kappa = 2$, 0.974 for $\kappa = 3$, 0.998 for $\kappa = 4$ and 0.999993 for $\kappa = 6$: almost everything sits just below 1, and the slopes the robot follows become tiny.
- **Only disks.** The formula holds for worlds of disks; other shapes must first be bent smoothly into a world of disks, which Rimon and Koditschek did for star-shaped obstacles. In a high-dimensional C-space the obstacles are known only through collision checks, so building such a bending is hard (LaValle §8.4.4.3; Choset slides).

8. Summary

Field	How it is built	From under the table	Cost
Attractive potential	half the squared distance to the door	straight into the table	one subtraction per step
Potential field (bowl + hills)	bowl plus a hill within 0.5 m of each obstacle	stops in a local minimum at (3.60, 0.59)	9 clearances per step
Randomized potential field	descend, random-walk when stuck, backtrack	door, 46.1 m driven (median of 200)	many tuned settings
Harmonic potential	obstacles 1, door 0, every point the average of its neighbours	door, 5.61 m, no local minimum	29.2 million updates on a grid
Navigation function (Rimon–Koditschek)	one formula on a world of disks, knob κ	door for $\kappa = 2$; a local minimum for $\kappa = 1$	disks only; flat for large κ

- The robot picks each step from where it stands by stepping down the **potential field** (G-2740), so it needs no grid: from (2, 1) it reaches the door after 3.47 m; the robot under the table stops at (3.60, 0.59), 3.41 m from the door, because there the door's pull up, 3.407, plus the wall's push, 0.155, balances the table's push down, 3.562.
- The **attractive potential** (G-2741), half the squared distance to the door, has minus its gradient pointing straight at the door, so alone it drives a straight line, the 3.4 m drive from (2, 1).
- The **repulsive potential** (G-2742) is zero beyond 0.5 m of clearance and grows without bound at contact, so the robot never touches an obstacle: the green robot passes 0.278 m from the table instead of 0.132 m.
- A **local minimum** (G-1110) appears where the pull meets an obstacle's push head on; the starts that roll into it fill the strip under the table, 7.7 percent of the starts, so gradient descent alone fails there every time.
- A **randomized potential field** (G-2744) escapes by a **random walk** (G-2745) until the height drops below the local minimum's, so it always reached the door in 200 runs, but drove a median 46.1 m against the shortest 4.55 m, because random steps wander back and forth.
- A **harmonic potential** (G-2748) obeys **Laplace's equation** (G-2746), so every free point is the average of its neighbours and none can be a local minimum; the robot under the table reaches the door in 5.61 m, after 5,000 sweeps of averaging over the whole grid.
- A **navigation function** (G-2750) has one minimum and the same top value on every boundary, so descent from almost every start reaches the goal; on a world of disks Rimon and Koditschek's formula gives one once κ is large enough, here from $\kappa = 2$.

The robot can now reach the door from under the table, by luck, on a grid, or in a world

of disks. Still missing: a planner that needs no grid, no tuned luck and no special shapes, and that works with only the collision checks and nearest neighbours of chapter RO-04; rapidly-exploring random trees grow a tree of random samples from the start instead.

9. Sources

Built from

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- Lynch, K. M. and Park, F. C. (2017). *Modern Robotics: Mechanics, Planning, and Control*. Cambridge University Press. §10.6.1 "A Point in C-space" and §10.6.2 "Navigation Functions". Free preprint: http://modernrobotics.org (MR §10.6.1, §10.6.2)
- Choset, H. (Carnegie Mellon University), *16-735 Robot Motion Planning*, slides "Potential Functions": attractive and repulsive potentials, gradient descent, navigation functions, sphere world (a world of disks), https://www.cs.cmu.edu/~motionplanning/lecture/Chap4-Potential-Field_howie.pdf (Choset slides)
- LaValle, S. M. (2006). *Planning Algorithms*. Cambridge University Press. §5.4.3 "Randomized Potential Fields", https://lavalle.pl/planning/node225.html; §8.4.1.2 "Navigation functions", https://lavalle.pl/planning/node391.html; §8.4.4.3 "Navigation function in the sense of Rimon-Koditschek", https://lavalle.pl/planning/node397.html; §8.4.4.4 "Harmonic potential functions", https://lavalle.pl/planning/node398.html (LaValle §5.4.3, §8.4.1.2, §8.4.4.3, §8.4.4.4)
- Wikipedia, "Harmonic function" (the mean value property and the maximum principle), https://en.wikipedia.org/wiki/Harmonic_function, and "Discrete Laplace operator" (the five-point stencil), https://en.wikipedia.org/wiki/Discrete_Laplace_operator (Wikipedia)

Other references

- Rimon, E. and Koditschek, D. E. (1992). "Exact robot navigation using artificial potential functions". *IEEE Transactions on Robotics and Automation* 8(5), 501–518, as presented in the Choset slides and LaValle §8.4.4.3

10. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Potential field (artificial potential field) (G-2740)	A height over the robot's C-space that is lowest at the goal and rises near obstacles, built as an attractive bowl plus a repulsive hill round each obstacle; the robot reaches the goal by stepping downhill from wherever it is, with no grid and no search, but can stop in a local minimum.
Attractive potential (G-2741)	The part of a potential field that pulls the robot to the goal: half the squared distance to the goal times a gain, $\frac{1}{2}\zeta\|q - q_{\text{goal}}\|^2$; minus its gradient points straight at the goal, like a spring.
Repulsive potential (G-2742)	The part of a potential field that pushes the robot away from one obstacle: zero beyond an influence distance, then growing without bound as the clearance shrinks, $\frac{1}{2}\eta(1/d - 1/Q^*)^2$; it keeps the robot from ever reaching the obstacle.
Randomized potential field (G-2744)	A planner that follows a potential field downhill and, when stuck in a local minimum, takes random walks until it finds a lower point, then descends again, backtracking after many failed walks; it escapes local minima, at the price of long paths and many tuned settings.
Random walk (G-2745)	A path made of steps each chosen at random, with no memory of the steps before, such as moving x and y by 0.05 m up or down by coin tosses; it wanders about the square root of the number of steps times the step size.
Harmonic potential (G-2748)	A potential field that is a harmonic function, held at 1 on every obstacle and 0 at the goal; it has no local minimum, so stepping downhill always reaches the goal, but it must be solved on a grid of the whole free space first.
Laplace's equation (G-2746)	The equation that a function's bend along x plus its bend along y is zero everywhere inside a region, $\partial^2\phi/\partial x^2 + \partial^2\phi/\partial y^2 = 0$; on a grid it says each point is the average of its four neighbours, so its solutions have no local minima inside.

Term	Meaning
Harmonic function (G-2747)	A function that obeys Laplace's equation, such as a settled temperature; its highest and lowest values lie on the boundary of its region, never inside, which is why it can guide a robot without local minima.
Navigation function (Rimon–Koditschek) (G-2750)	A smooth field on free space with only one minimum, at the goal, the same top value on the whole boundary of free space and no flat stationary points; gradient descent on it reaches the goal from almost every start. Rimon and Koditschek gave a formula for worlds of disks, a disk-shaped free space with disk-shaped obstacles.
Wavefront propagation (wavefront planner) (G-2700)	Numbering the free cells of a grid ring by ring outwards from the goal, each ring one move further, so every cell gets its fewest moves to the goal in one pass (8 for the room's start cell); a breadth-first search from the goal, used when every move costs the same.
D* (dynamic A*) (G-2713)	A replanner that keeps the cost-to-go from a backward search and, when sensed costs differ from the map, puts the affected cells back in a priority queue and fixes them until the robot's own cost-to-go is right again; it repairs only the numbers a change affects instead of searching from scratch.
Contour plot (G-468)	A map of a surface seen from above, with lines joining points of equal height, so a 3D shape such as a loss bowl can be drawn and read on flat paper.
Configuration (G-2286)	The list of numbers that fixes the position of every point of a robot, such as $q = (x, y, \theta) = (2, 1, 0.524)$ for a floor robot or the joint angles (θ_1, θ_2) of a 2R arm; with it a planner treats the whole robot as one point.
Gradient (of the loss) (G-863)	How fast the loss changes with each weight and bias, collected in one vector (the partial derivatives of the loss); it points in the direction in which the loss rises fastest.
Gradient descent (G-862)	Finding the lowest point of a function by repeated small steps downhill.

Term	Meaning
Learning rate (η) (G-1068)	How strongly each update changes the model; in gradient descent, the number the slope is multiplied by to get the step size.
Distance query (G-2674)	A geometric query that returns the shortest distance between two bodies, 0 if they overlap; it tells a planner how much room the robot has, not only whether it is free.
Local minimum (G-1110)	A point lower than everything around it, but not the lowest overall.
Global minimum (G-848)	The lowest point of the whole function.
Stationary point (G-1879)	A point where a function is flat, its slope is zero (the derivative, or every partial derivative, is zero); it is a minimum, a maximum or a saddle point.
Saddle point (G-1718)	A flat point that curves up in one direction and down in another.
Value iteration (G-2704)	Computing every state's cost-to-go by repeating, over all states, the update "value = smallest move cost plus the next state's value" until no value changes; it needs no priority queue and no order, and it carries over to values between grid cells and to uncertain moves.
Navigation function (G-2702)	A number for every state that is 0 at the goal, infinite only where the goal cannot be reached, and lower at some neighbour of every other state, so it has one minimum, the goal; stepping to the neighbour with the smallest move plus number from any state leads to the goal, by the shortest route when the numbers are the cost-to-go.